

Table 6: Simulation IV: Empirical power and Monte Carlo standard error for $k = 1$, under different values of c .

c	n	$r=2$			$r=5$			$r=8$		
		Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
0.5	50	0.0000 (0.0000)	0.0452 (0.0029)	0.0510 (0.0031)	0.0000 (0.0000)	0.0452 (0.0029)	0.0430 (0.0029)	0.0000 (0.0000)	0.0550 (0.0032)	0.0448 (0.0029)
	100	0.0000 (0.0000)	0.0500 (0.0031)	0.0422 (0.0028)	0.0000 (0.0000)	0.0534 (0.0032)	0.0554 (0.0032)	0.0000 (0.0000)	0.0520 (0.0031)	0.0426 (0.0029)
	200	0.0000 (0.0000)	0.0518 (0.0031)	0.0494 (0.0031)	0.0000 (0.0000)	0.0494 (0.0031)	0.0484 (0.0030)	0.0000 (0.0000)	0.0528 (0.0032)	0.0436 (0.0029)
	500	0.0000 (0.0000)	0.0440 (0.0029)	0.0500 (0.0031)	0.0000 (0.0000)	0.0498 (0.0031)	0.0472 (0.0030)	0.0000 (0.0000)	0.0504 (0.0031)	0.0430 (0.0029)
	1000	0.0000 (0.0000)	0.0454 (0.0029)	0.0480 (0.0030)	0.0000 (0.0000)	0.0460 (0.0030)	0.0472 (0.0030)	0.0000 (0.0000)	0.0480 (0.0030)	0.0536 (0.0032)
1	50	0.0000 (0.0000)	0.0516 (0.0031)	0.0520 (0.0031)	0.0000 (0.0000)	0.0478 (0.0030)	0.0414 (0.0028)	0.0000 (0.0000)	0.0556 (0.0032)	0.0460 (0.0030)
	100	0.0000 (0.0000)	0.0480 (0.0030)	0.0392 (0.0027)	0.0000 (0.0000)	0.0492 (0.0031)	0.0550 (0.0032)	0.0000 (0.0000)	0.0476 (0.0030)	0.0458 (0.0030)
	200	0.0000 (0.0000)	0.0558 (0.0032)	0.0524 (0.0032)	0.0000 (0.0000)	0.0536 (0.0032)	0.0466 (0.0030)	0.0000 (0.0000)	0.0470 (0.0030)	0.0526 (0.0032)
	500	0.0000 (0.0000)	0.0508 (0.0031)	0.0484 (0.0030)	0.0000 (0.0000)	0.0492 (0.0031)	0.0434 (0.0029)	0.0000 (0.0000)	0.0530 (0.0032)	0.0474 (0.0030)
	1000	0.0000 (0.0000)	0.0470 (0.0030)	0.0432 (0.0029)	0.0000 (0.0000)	0.0414 (0.0028)	0.0408 (0.0028)	0.0000 (0.0000)	0.0462 (0.0030)	0.0492 (0.0031)
1.5	50	0.0000 (0.0000)	0.0448 (0.0029)	0.0560 (0.0033)	0.0000 (0.0000)	0.0462 (0.0030)	0.0456 (0.0030)	0.0000 (0.0000)	0.0522 (0.0031)	0.0394 (0.0028)
	100	0.0000 (0.0000)	0.0486 (0.0030)	0.0458 (0.0030)	0.0000 (0.0000)	0.0464 (0.0030)	0.0614 (0.0034)	0.0000 (0.0000)	0.0496 (0.0031)	0.0438 (0.0029)
	200	0.0000 (0.0000)	0.0558 (0.0032)	0.0488 (0.0030)	0.0000 (0.0000)	0.0522 (0.0031)	0.0400 (0.0028)	0.0000 (0.0000)	0.0564 (0.0033)	0.0536 (0.0032)
	500	0.0000 (0.0000)	0.0466 (0.0030)	0.0452 (0.0029)	0.0000 (0.0000)	0.0532 (0.0032)	0.0474 (0.0030)	0.0000 (0.0000)	0.0506 (0.0031)	0.0442 (0.0029)
	1000	0.0000 (0.0000)	0.0442 (0.0029)	0.0436 (0.0029)	0.0000 (0.0000)	0.0434 (0.0029)	0.0578 (0.0033)	0.0000 (0.0000)	0.0456 (0.0030)	0.0514 (0.0031)
2	50	0.0000 (0.0000)	0.0488 (0.0030)	0.0484 (0.0030)	0.0000 (0.0000)	0.0472 (0.0030)	0.0344 (0.0026)	0.0000 (0.0000)	0.0428 (0.0029)	0.0384 (0.0027)
	100	0.0000 (0.0000)	0.0454 (0.0029)	0.0410 (0.0028)	0.0000 (0.0000)	0.0502 (0.0031)	0.0512 (0.0031)	0.0000 (0.0000)	0.0456 (0.0030)	0.0430 (0.0029)
	200	0.0000 (0.0000)	0.0532 (0.0032)	0.0484 (0.0030)	0.0000 (0.0000)	0.0568 (0.0033)	0.0432 (0.0029)	0.0000 (0.0000)	0.0454 (0.0029)	0.0388 (0.0027)
	500	0.0000 (0.0000)	0.0526 (0.0032)	0.0506 (0.0031)	0.0000 (0.0000)	0.0544 (0.0032)	0.0416 (0.0028)	0.0000 (0.0000)	0.0548 (0.0032)	0.0398 (0.0028)
	1000	0.0000 (0.0000)	0.0478 (0.0030)	0.0420 (0.0028)	0.0000 (0.0000)	0.0480 (0.0030)	0.0486 (0.0030)	0.0000 (0.0000)	0.0450 (0.0029)	0.0498 (0.0031)

Table 7: Simulation IV: Empirical power and Monte Carlo standard error for $k = 2$, under different values of c .

c	n	$r=2$			$r=5$			$r=8$		
		Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
0.5	50	0.0000 (0.0000)	0.0494 (0.0031)	0.0534 (0.0032)	0.0000 (0.0000)	0.0618 (0.0034)	0.0406 (0.0028)	0.0000 (0.0000)	0.0552 (0.0032)	0.0508 (0.0031)
	100	0.0000 (0.0000)	0.0510 (0.0031)	0.0464 (0.0030)	0.0000 (0.0000)	0.0518 (0.0031)	0.0530 (0.0032)	0.0000 (0.0000)	0.0520 (0.0031)	0.0482 (0.0030)
	200	0.0000 (0.0000)	0.0490 (0.0031)	0.0532 (0.0032)	0.0000 (0.0000)	0.0532 (0.0032)	0.0428 (0.0029)	0.0000 (0.0000)	0.0514 (0.0031)	0.0460 (0.0030)
	500	0.0000 (0.0000)	0.0530 (0.0032)	0.0522 (0.0031)	0.0000 (0.0000)	0.0534 (0.0032)	0.0428 (0.0029)	0.0000 (0.0000)	0.0540 (0.0032)	0.0448 (0.0029)
	1000	0.0000 (0.0000)	0.0434 (0.0029)	0.0476 (0.0030)	0.0000 (0.0000)	0.0438 (0.0029)	0.0546 (0.0032)	0.0000 (0.0000)	0.0504 (0.0031)	0.0506 (0.0031)
1	50	0.0000 (0.0000)	0.0452 (0.0029)	0.0520 (0.0031)	0.0000 (0.0000)	0.0478 (0.0030)	0.0472 (0.0030)	0.0000 (0.0000)	0.0578 (0.0033)	0.0514 (0.0031)
	100	0.0000 (0.0000)	0.0460 (0.0030)	0.0452 (0.0029)	0.0000 (0.0000)	0.0574 (0.0033)	0.0600 (0.0034)	0.0000 (0.0000)	0.0490 (0.0031)	0.0548 (0.0032)
	200	0.0000 (0.0000)	0.0476 (0.0030)	0.0482 (0.0030)	0.0000 (0.0000)	0.0548 (0.0032)	0.0434 (0.0029)	0.0000 (0.0000)	0.0518 (0.0031)	0.0506 (0.0031)
	500	0.0000 (0.0000)	0.0526 (0.0032)	0.0494 (0.0031)	0.0000 (0.0000)	0.0540 (0.0032)	0.0472 (0.0030)	0.0000 (0.0000)	0.0562 (0.0033)	0.0424 (0.0028)
	1000	0.0000 (0.0000)	0.0428 (0.0029)	0.0484 (0.0030)	0.0000 (0.0000)	0.0452 (0.0029)	0.0566 (0.0033)	0.0000 (0.0000)	0.0528 (0.0032)	0.0538 (0.0032)
1.5	50	0.0000 (0.0000)	0.0510 (0.0031)	0.0584 (0.0033)	0.0000 (0.0000)	0.0530 (0.0032)	0.0508 (0.0031)	0.0000 (0.0000)	0.0664 (0.0035)	0.0540 (0.0032)
	100	0.0000 (0.0000)	0.0482 (0.0030)	0.0458 (0.0030)	0.0000 (0.0000)	0.0604 (0.0034)	0.0626 (0.0034)	0.0000 (0.0000)	0.0596 (0.0033)	0.0588 (0.0033)
	200	0.0000 (0.0000)	0.0544 (0.0032)	0.0590 (0.0033)	0.0000 (0.0000)	0.0562 (0.0033)	0.0406 (0.0028)	0.0000 (0.0000)	0.0510 (0.0031)	0.0518 (0.0031)
	500	0.0000 (0.0000)	0.0514 (0.0031)	0.0450 (0.0029)	0.0000 (0.0000)	0.0512 (0.0031)	0.0474 (0.0030)	0.0000 (0.0000)	0.0564 (0.0033)	0.0450 (0.0029)
	1000	0.0000 (0.0000)	0.0464 (0.0030)	0.0464 (0.0030)	0.0000 (0.0000)	0.0426 (0.0029)	0.0668 (0.0035)	0.0000 (0.0000)	0.0500 (0.0031)	0.0568 (0.0033)
2	50	0.0000 (0.0000)	0.0544 (0.0032)	0.0556 (0.0032)	0.0000 (0.0000)	0.0662 (0.0035)	0.0520 (0.0031)	0.0000 (0.0000)	0.0708 (0.0036)	0.0538 (0.0032)
	100	0.0000 (0.0000)	0.0530 (0.0032)	0.0498 (0.0031)	0.0000 (0.0000)	0.0614 (0.0037)	0.0724 (0.0037)	0.0000 (0.0000)	0.0620 (0.0034)	0.0626 (0.0034)
	200	0.0000 (0.0000)	0.0550 (0.0032)	0.0570 (0.0033)	0.0000 (0.0000)	0.0652 (0.0035)	0.0476 (0.0030)	0.0000 (0.0000)	0.0594 (0.0033)	0.0550 (0.0032)
	500	0.0000 (0.0000)	0.0534 (0.0032)	0.0526 (0.0032)	0.0000 (0.0000)	0.0564 (0.0033)	0.0468 (0.0030)	0.0000 (0.0000)	0.0576 (0.0033)	0.0418 (0.0028)
	1000	0.0000 (0.0000)	0.0440 (0.0029)	0.0436 (0.0029)	0.0000 (0.0000)	0.0448 (0.0029)	0.0500 (0.0031)	0.0000 (0.0000)	0.0508 (0.0031)	0.0514 (0.0031)

k = 4										
c	n	r=2			r=5			r=8		
		Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
0.5	50	0.0000 (0.0000)	0.0424 (0.0028)	0.0558 (0.0032)	0.0000 (0.0000)	0.0490 (0.0031)	0.0430 (0.0029)	0.0000 (0.0000)	0.0496 (0.0031)	0.0540 (0.0032)
	100	0.0000 (0.0000)	0.0488 (0.0030)	0.0436 (0.0029)	0.0000 (0.0000)	0.0524 (0.0032)	0.0504 (0.0031)	0.0000 (0.0000)	0.0508 (0.0031)	0.0466 (0.0030)
	200	0.0000 (0.0000)	0.0534 (0.0032)	0.0550 (0.0032)	0.0000 (0.0000)	0.0478 (0.0030)	0.0470 (0.0030)	0.0000 (0.0000)	0.0536 (0.0032)	0.0544 (0.0032)
	500	0.0000 (0.0000)	0.0474 (0.0030)	0.0486 (0.0030)	0.0000 (0.0000)	0.0518 (0.0031)	0.0462 (0.0030)	0.0000 (0.0000)	0.0542 (0.0032)	0.0420 (0.0028)
	1000	0.0000 (0.0000)	0.0414 (0.0028)	0.0460 (0.0030)	0.0000 (0.0000)	0.0440 (0.0029)	0.0496 (0.0031)	0.0000 (0.0000)	0.0482 (0.0030)	0.0578 (0.0033)
1	50	0.0000 (0.0000)	0.0464 (0.0030)	0.0530 (0.0032)	0.0000 (0.0000)	0.0482 (0.0030)	0.0444 (0.0029)	0.0000 (0.0000)	0.0510 (0.0031)	0.0486 (0.0030)
	100	0.0000 (0.0000)	0.0512 (0.0031)	0.0424 (0.0028)	0.0000 (0.0000)	0.0504 (0.0031)	0.0576 (0.0033)	0.0000 (0.0000)	0.0520 (0.0031)	0.0446 (0.0029)
	200	0.0000 (0.0000)	0.0506 (0.0031)	0.0488 (0.0030)	0.0000 (0.0000)	0.0476 (0.0030)	0.0460 (0.0030)	0.0000 (0.0000)	0.0488 (0.0030)	0.0464 (0.0030)
	500	0.0000 (0.0000)	0.0506 (0.0031)	0.0520 (0.0031)	0.0000 (0.0000)	0.0562 (0.0033)	0.0484 (0.0030)	0.0000 (0.0000)	0.0500 (0.0031)	0.0464 (0.0030)
	1000	0.0000 (0.0000)	0.0454 (0.0029)	0.0460 (0.0030)	0.0000 (0.0000)	0.0406 (0.0028)	0.0570 (0.0033)	0.0000 (0.0000)	0.0498 (0.0031)	0.0580 (0.0033)
1.5	50	0.0000 (0.0000)	0.0468 (0.0030)	0.0514 (0.0031)	0.0000 (0.0000)	0.0450 (0.0029)	0.0466 (0.0030)	0.0000 (0.0000)	0.0556 (0.0032)	0.0506 (0.0031)
	100	0.0000 (0.0000)	0.0510 (0.0031)	0.0442 (0.0029)	0.0000 (0.0000)	0.0576 (0.0033)	0.0538 (0.0032)	0.0000 (0.0000)	0.0514 (0.0031)	0.0490 (0.0031)
	200	0.0000 (0.0000)	0.0538 (0.0032)	0.0480 (0.0030)	0.0000 (0.0000)	0.0550 (0.0032)	0.0500 (0.0031)	0.0000 (0.0000)	0.0492 (0.0031)	0.0534 (0.0032)
	500	0.0000 (0.0000)	0.0496 (0.0031)	0.0468 (0.0030)	0.0000 (0.0000)	0.0526 (0.0032)	0.0454 (0.0029)	0.0000 (0.0000)	0.0482 (0.0030)	0.0412 (0.0028)
	1000	0.0000 (0.0000)	0.0452 (0.0029)	0.0478 (0.0030)	0.0000 (0.0000)	0.0482 (0.0030)	0.0586 (0.0033)	0.0000 (0.0000)	0.0518 (0.0031)	0.0610 (0.0034)
2	50	0.0000 (0.0000)	0.0502 (0.0031)	0.0584 (0.0033)	0.0000 (0.0000)	0.0466 (0.0030)	0.0444 (0.0029)	0.0000 (0.0000)	0.0448 (0.0029)	0.0610 (0.0034)
	100	0.0000 (0.0000)	0.0562 (0.0033)	0.0498 (0.0031)	0.0000 (0.0000)	0.0618 (0.0034)	0.0616 (0.0034)	0.0000 (0.0000)	0.0498 (0.0031)	0.0504 (0.0031)
	200	0.0000 (0.0000)	0.0550 (0.0032)	0.0572 (0.0033)	0.0000 (0.0000)	0.0572 (0.0033)	0.0544 (0.0032)	0.0000 (0.0000)	0.0592 (0.0033)	0.0552 (0.0032)
	500	0.0000 (0.0000)	0.0516 (0.0031)	0.0502 (0.0031)	0.0000 (0.0000)	0.0560 (0.0033)	0.0504 (0.0031)	0.0000 (0.0000)	0.0518 (0.0031)	0.0444 (0.0029)
	1000	0.0000 (0.0000)	0.0474 (0.0030)	0.0500 (0.0031)	0.0000 (0.0000)	0.0440 (0.0029)	0.0546 (0.0032)	0.0000 (0.0000)	0.0520 (0.0031)	0.0490 (0.0031)
k = 8										
c	n	r=2			r=5			r=8		
		Uniform	Central	Tail	Uniform	Central	Tail	Uniform	Central	Tail
0.5	50	0.0000 (0.0000)	0.0466 (0.0030)	0.0568 (0.0033)	0.0000 (0.0000)	0.0484 (0.0030)	0.0476 (0.0030)	0.0000 (0.0000)	0.0548 (0.0032)	0.0502 (0.0031)
	100	0.0000 (0.0000)	0.0546 (0.0032)	0.0446 (0.0029)	0.0000 (0.0000)	0.0542 (0.0032)	0.0552 (0.0032)	0.0000 (0.0000)	0.0492 (0.0031)	0.0476 (0.0030)
	200	0.0000 (0.0000)	0.0536 (0.0032)	0.0560 (0.0033)	0.0000 (0.0000)	0.0476 (0.0030)	0.0464 (0.0030)	0.0000 (0.0000)	0.0518 (0.0031)	0.0446 (0.0029)
	500	0.0000 (0.0000)	0.0500 (0.0031)	0.0554 (0.0032)	0.0000 (0.0000)	0.0504 (0.0031)	0.0446 (0.0029)	0.0000 (0.0000)	0.0538 (0.0032)	0.0384 (0.0027)
	1000	0.0000 (0.0000)	0.0480 (0.0030)	0.0492 (0.0031)	0.0000 (0.0000)	0.0422 (0.0028)	0.0496 (0.0031)	0.0000 (0.0000)	0.0470 (0.0030)	0.0578 (0.0033)
1	50	0.0000 (0.0000)	0.0468 (0.0030)	0.0556 (0.0032)	0.0000 (0.0000)	0.0476 (0.0030)	0.0458 (0.0030)	0.0000 (0.0000)	0.0458 (0.0030)	0.0470 (0.0030)
	100	0.0000 (0.0000)	0.0530 (0.0032)	0.0460 (0.0030)	0.0000 (0.0000)	0.0560 (0.0033)	0.0532 (0.0032)	0.0000 (0.0000)	0.0502 (0.0031)	0.0504 (0.0031)
	200	0.0000 (0.0000)	0.0484 (0.0030)	0.0522 (0.0031)	0.0000 (0.0000)	0.0510 (0.0031)	0.0438 (0.0029)	0.0000 (0.0000)	0.0508 (0.0031)	0.0472 (0.0030)
	500	0.0000 (0.0000)	0.0496 (0.0031)	0.0502 (0.0031)	0.0000 (0.0000)	0.0576 (0.0033)	0.0488 (0.0030)	0.0000 (0.0000)	0.0474 (0.0030)	0.0446 (0.0029)
	1000	0.0000 (0.0000)	0.0426 (0.0029)	0.0480 (0.0030)	0.0000 (0.0000)	0.0446 (0.0029)	0.0558 (0.0032)	0.0000 (0.0000)	0.0462 (0.0030)	0.0508 (0.0031)
1.5	50	0.0000 (0.0000)	0.0478 (0.0030)	0.0520 (0.0031)	0.0000 (0.0000)	0.0466 (0.0030)	0.0542 (0.0032)	0.0000 (0.0000)	0.0530 (0.0032)	0.0450 (0.0029)
	100	0.0000 (0.0000)	0.0478 (0.0030)	0.0436 (0.0029)	0.0000 (0.0000)	0.0494 (0.0031)	0.0522 (0.0031)	0.0000 (0.0000)	0.0464 (0.0030)	0.0470 (0.0030)
	200	0.0000 (0.0000)	0.0538 (0.0032)	0.0516 (0.0031)	0.0000 (0.0000)	0.0510 (0.0031)	0.0452 (0.0029)	0.0000 (0.0000)	0.0508 (0.0031)	0.0528 (0.0032)
	500	0.0000 (0.0000)	0.0468 (0.0030)	0.0472 (0.0030)	0.0000 (0.0000)	0.0522 (0.0031)	0.0442 (0.0029)	0.0000 (0.0000)	0.0526 (0.0032)	0.0426 (0.0029)
	1000	0.0000 (0.0000)	0.0460 (0.0030)	0.0516 (0.0031)	0.0000 (0.0000)	0.0452 (0.0029)	0.0570 (0.0033)	0.0000 (0.0000)	0.0482 (0.0030)	0.0474 (0.0030)
2	50	0.0000 (0.0000)	0.0444 (0.0029)	0.0440 (0.0029)	0.0000 (0.0000)	0.0542 (0.0032)	0.0508 (0.0031)	0.0000 (0.0000)	0.0472 (0.0030)	0.0526 (0.0032)
	100	0.0000 (0.0000)	0.0470 (0.0030)	0.0416 (0.0028)	0.0000 (0.0000)	0.0470 (0.0030)	0.0456 (0.0030)	0.0000 (0.0000)	0.0462 (0.0030)	0.0454 (0.0029)
	200	0.0000 (0.0000)	0.0528 (0.0032)	0.0540 (0.0032)	0.0000 (0.0000)	0.0546 (0.0032)	0.0490 (0.0031)	0.0000 (0.0000)	0.0548 (0.0032)	0.0502 (0.0031)
	500	0.0000 (0.0000)	0.0526 (0.0032)	0.0484 (0.0030)	0.0000 (0.0000)	0.0570 (0.0033)	0.0474 (0.0030)	0.0000 (0.0000)	0.0542 (0.0032)	0.0402 (0.0028)
	1000	0.0000 (0.0000)	0.0498 (0.0031)	0.0476 (0.0030)	0.0000 (0.0000)	0.0474 (0.0030)	0.0564 (0.0033)	0.0000 (0.0000)	0.0524 (0.0032)	0.0518 (0.0031)

Table 8: Simulation IV: Empirical power and Monte Carlo standard error for $k = 4$ and $k = 8$, under different values of c , retaining $r = 2, 5, 8$.